CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 7: Indices (Exponents)

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Evaluate $(32)^{\frac{3}{5}}$ and $(81)^{-\frac{1}{4}}$.
2. Evaluate $\left(\frac{16}{81}\right)^{-\frac{3}{4}}$.
3. Simplify $\frac{3^7 \times 9^{-2}}{27}$.
4. Simplify $(a^{\frac{1}{2}} + b^{\frac{1}{2}})(a^{\frac{1}{2}} - b^{\frac{1}{2}})$.
5. Simplify $\sqrt{x^{-2}y^4} \times x^3 y^{-1}$.
6. Solve $3^{x-1} = 81$.
7. Solve $4^x = 8^{x-2}$.
8. Solve $5^{2x+1} = 125^{x-1}$.
9. Solve $3^{x+1} + 3^x = 108$.
10. Solve $2^{2x} - 6 \times 2^x + 8 = 0$.

MathSetu

Answer Key

1. 8 and $\frac{1}{3}$
2. $\frac{27}{8}$
3. $3^0 = 1$
4. $a - b$
5. x^2y
6. $x = 5$
7. $x = 6$
8. $x = 4$
9. $x = 3$
10. $x = 1$ or $x = 2$

MathSetu